**The due date for submitting this assignment has passed.**
**Due on 2026-02-25, 23:59 IST.**

**You may submit any number of times before the due date. The final submission will be considered for grading.**

You have last submitted on: 2026-02-25, 08:53 IST

**Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.**

**Multiple Select Questions (MSQ)**

1) In a particular year, the profit (in lakhs of ₹) of Star Fish company is given by the polynomial $P(x) = ax^2 + bx + c$ where $x$ denotes the number of months since the beginning of the year (i.e., $x = 1$ denotes January, $x = 2$ denotes February, and so on). In January and February the company made a loss of ₹45(in lakhs), and ₹19(in lakhs) respectively, and in March the company made a profit of ₹3(in lakhs). Let the loss be represented by negative of profit. Choose the correct set of options based on the given information. ***1 point***

- [ ] The maximum profit will be in the month of May.
- [x] The maximum profit will be in the month of August.
- [x] The maximum monthly profit amount is ₹53 lakh.
- [ ] The maximum monthly profit amount is ₹35 lakh.

Yes, the answer is correct.
Score: 1

Accepted Answers:

The maximum profit will be in the month of August.
The maximum monthly profit amount is ₹53 lakh.

2) If $A$ be a $3 \times 4$ matrix and $b$ be a $3 \times 1$ matrix, then choose the set of correct options. ***1 point***

- [x] If $(A \mid b)$ be the augmented matrix and $(A' \mid b')$ be the matrix obtained from $(A \mid b)$ after a finite number of elementary row operations then the system $Ax = b$ and the system $A'x = b'$ have the same set of solutions.

- [ ] If $(A^{'}|b^{'})$ is the reduced row echelon form of $(A|b)$ then the system $A^{'}x = b^{'}$ has at least one solution.
- [x] If $(A^{'}|b^{'})$ is the reduced row echelon form of $(A|b)$, then $A^{'}$ is also in reduced row echelon form.
- [x] If $(A^{'}|b^{'})$ is the reduced row echelon form of $(A|b)$ and there is no row such that the only non zero entry lies in the last column of $(A^{'}|b^{'})$ then the system $Ax = b$ has at least one solution.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If $(A|b)$ be the augmented matrix and $(A^{'}|b^{'})$ be the matrix obtained from $(A|b)$ after a finite number of elementary row operations then the system $Ax = b$ and the system $A^{'}x = b^{'}$ have the same set of solutions.
If $(A^{'}|b^{'})$ is the reduced row echelon form of $(A|b)$, then $A^{'}$ is also in reduced row echelon form.
If $(A^{'}|b^{'})$ is the reduced row echelon form of $(A|b)$ and there is no row such that the only non zero entry lies in the last column of $(A^{'}|b^{'})$ then the system $Ax = b$ has at least one solution.

3) Choose the set of correct options ***1 point***

- [x] If the sum of all the elements of each row of a matrix $A$ is 0, then $A$ is not invertible.
- [x] If $E$ is a matrix of order $3 \times 3$ obtained from the identity matrix by a finite number of elementary row operations then $E$ is invertible.
- [ ] Any system of linear equations has at least one solution.
- [ ] If $A$ is a matrix of order $3 \times 3$ and $det(A) = 3$ then $det(Adj(A)) = 3$.
- [x] If $A$ is a matrix of order $3 \times 3$ and $det(A) = 3$ then $det(Adj(A)) = 9$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

If the sum of all the elements of each row of a matrix $A$ is 0, then $A$ is not invertible.
If $E$ is a matrix of order $3 \times 3$ obtained from the identity matrix by a finite number of elementary row operations then $E$ is invertible.
If $A$ is a matrix of order $3 \times 3$ and $det(A) = 3$ then $det(Adj(A)) = 9$.

4) Ramya bought 1 comic book, 2 horror books, and 1 novel from a bookshop which cost her ₹1000. Romy bought 2 comic books, 5 horror books, and 1 novel which cost him ₹2000. Farjana bought 4 comic books, 5 horror books, and $c$ novels from a shop which cost her ₹$d$. If $x_1, x_2,$ and $x_3$ represent the price of each comic book, horror book, and novel, respectively, then choose the set of correct options. ***1 point***

- [x] The matrix representation to find $x_1, x_2$ and $x_3$ is $\begin{bmatrix}1 & 2 & 1\\2 & 5 & 1\\4 & 5 & c\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}1000\\2000\\d\end{bmatrix}$
- [x] The matrix representation to find $x_1, x_2$ and $x_3$ is $[x_1 \quad x_2 \quad x_3]\begin{bmatrix}1 & 2 & 4\\2 & 5 & 5\\1 & 1 & c\end{bmatrix} = [1000 \quad 2000 \quad d]$
- [ ] The matrix representation to find $x_1, x_2$ and $x_3$ is $\begin{bmatrix}1 & 2 & 4\\2 & 5 & 5\\1 & 1 & c\end{bmatrix}[x_1 \quad x_2 \quad x_3] = [1000 \quad 2000 \quad d]$
- [ ] If Farjana tries to find $x_1, x_2,$ and $x_3$ using appropriate matrix representation by taking $c = 2$ and $d = 4000$, then the price of each comic book that she thus arrives at, will not be unique.
- [ ] If $c = 7$ and $d = 4000$, then the price of each comic book cannot be determined from this data.
- [x] If $c = 7$ and $d = 3000$, then the shopkeeper has made a mistake.
- [x] If $c = 2$ and $d = 3000$, then the price of each comic book can be determined from the data.

Partially Correct.

Score: 0.8

Accepted Answers:

The matrix representation to find $x_1, x_2$ and $x_3$ is $\begin{bmatrix}1 & 2 & 1\\2 & 5 & 1\\4 & 5 & c\end{bmatrix}\begin{bmatrix}x_1\\x_2\\x_3\end{bmatrix} = \begin{bmatrix}1000\\2000\\d\end{bmatrix}$

The matrix representation to find $x_1, x_2$ and $x_3$ is $[x_1 \quad x_2 \quad x_3]\begin{bmatrix}1 & 2 & 4\\2 & 5 & 5\\1 & 1 & c\end{bmatrix} = [1000 \quad 2000 \quad d]$

If $c = 7$ and $d = 4000$, then the price of each comic book cannot be determined from this data.

If $c = 7$ and $d = 3000$, then the shopkeeper has made a mistake.

If $c = 2$ and $d = 3000$, then the price of each comic book can be determined from the data.

5) Let A be an m × n matrix such that m < n. How many solutions does Ax = 0 have? ***1 point***

- Exactly one solution.
- Infinitely many solutions.
- No solution.
- Insufficient data. This depends on the entries of A.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Infinitely many solutions.

6) Let $A$ be an $n \times n$ matrix such that $\sum_{j=1}^{n} a_{ij} = 0$ for all $i$. How many solutions does $A\mathbf{x} = \mathbf{0}$ have? ***1 point***

- Exactly one solution.
- Infinitely many solutions.
- No solution.
- Insufficient data. This depends on the entries of A.

Yes, the answer is correct.

Score: 1

Accepted Answers:

Infinitely many solutions.

**Numerical Answer Type (NAT):**

7) How many solutions does the following system have? Enter 1 if there is only one solution, 0 if there are no solutions and -1 if there are infinitely many solutions. The coefficient matrix $A$ and the vector $b$ are given below:

$$A = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 1 & -1 \\ 2 & 1 & -2 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$$

-1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -1

***1 point***

8) Let the reduced row echelon form of a matrix $A$ be $\begin{bmatrix} 1 & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & 0 & -\frac{1}{6} \\ 0 & 0 & 1 & \frac{1}{6} \end{bmatrix}$. The first, second, third and fourth columns of $A$ are $\begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$, $\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$, $\begin{bmatrix} a \\ b \\ c \end{bmatrix}$ and $\begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$, respectively. The value of $a + b + c$ is

-12

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

***1 point***

9) $A$ is the reduced row echelon form of the matrix $\begin{bmatrix} 1 & 3 & 0 & 0 \\ 4 & 1 & 5 & 5 \\ 2 & 2 & 7 & 91 \\ 3 & 9 & 0 & 0 \end{bmatrix}$. Then determinant of $A$ is

0

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

***1 point***

10) If $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a solution of the system of equations

$$\begin{aligned} 7x + 2y + z &= 8 \\ 3y - z &= 2 \\ -3x + 4y - 2z &= 5 \end{aligned}$$

, then the value of $x + y + z$ is

121

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 121

***1 point***

11) Let $A = [1\ 7\ 2\ 9]$ and $M$ denote the reduced row echelon form of $A^T A$. The number of non-zero rows of $M$ is

1

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 1

***1 point***

12) Consider the curve corresponding to the function $f(x) = ax^3 + bx^2 + cx + d$. The following points lie on the curve: $(1, 1), (2, 7), (-1, -5), (3, 23)$. Find the value of $a - b + c - d$.

5

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 5

***1 point***

**Comprehension Type Question:**

The network in Figure: M2W2GA1 shows a proposed plan for flow of traffic around a park. All the streets are assumed to be one-way and the arrows denote the direction of flow of traffic. The plan calls for a computerized traffic light at the South Street. Let $2x_1, 3x_2, 2x_3,$ and $x_4$ denote the average number (per hour) of vehicles expected to pass through the connecting streets (e.g., $2x_1$ denote the average number (per hour) of vehicles expected to pass through the street connecting the North Street and West Street as shown in Figure: M2W2GA1). 400, 1000, 900, and $c$ denote the average number (per hour) of vehicles expected to pass through West, North, East, and South Streets respectively.

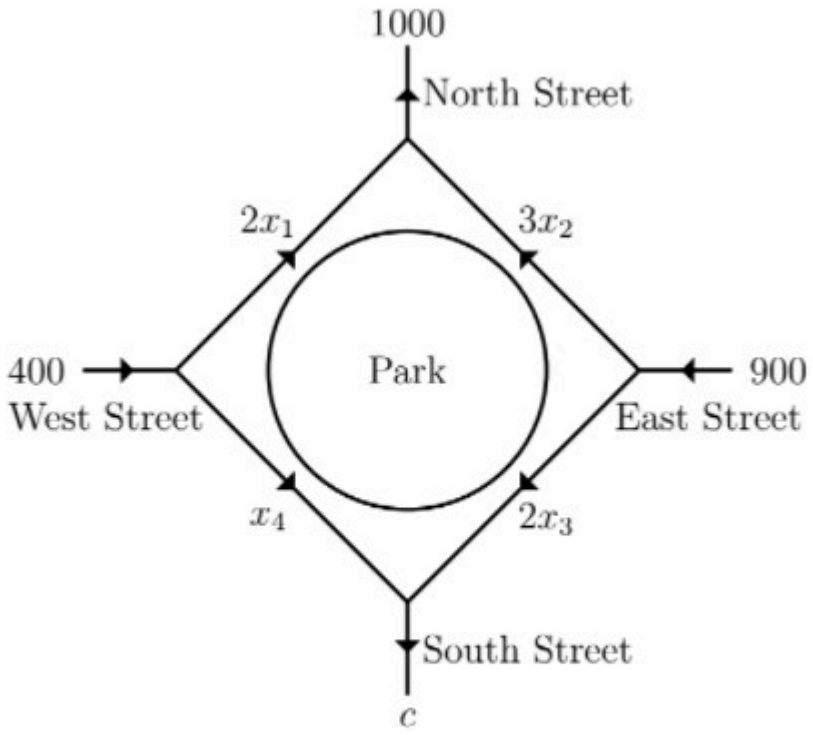



Figure: M2W2GA1

13) Which of the following options are correct? ***1 point***

The system of equations corresponding to the flow of expected traffic according to the given data above, will be

$$\begin{aligned} 2x_1 + 3x_2 &= 1000 \\ 3x_2 + 2x_3 &= 900 \\ 2x_3 + x_4 &= c \\ 2x_1 + x_4 &= 400 \end{aligned}$$

The system of equations corresponding to the flow of expected traffic according to the given data above, will be

$$\begin{aligned} 2x_1 + 3x_2 &= 900 \\ 3x_2 + 2x_3 &= 1000 \\ 2x_3 + x_4 &= 400 \\ 2x_1 + x_4 &= c \end{aligned}$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is

$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 900 \\ 1000 \\ 400 \\ c \end{bmatrix}$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is

$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1000 \\ 900 \\ c \\ 400 \end{bmatrix}$$

Partially Correct.

Score: 0.5

Accepted Answers:

The system of equations corresponding to the flow of expected traffic according to the given data above, will be

$$2x_1 + 3x_2 = 1000$$
$$3x_2 + 2x_3 = 900$$
$$2x_3 + x_4 = c$$
$$2x_1 + x_4 = 400$$

The matrix representation of the system of equations corresponding to the flow of expected traffic according to the given data above is

$$\begin{bmatrix} 2 & 3 & 0 & 0 \\ 0 & 3 & 2 & 0 \\ 0 & 0 & 2 & 1 \\ 2 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} 1000 \\ 900 \\ c \\ 400 \end{bmatrix}$$

14) How many vehicles are expected to pass through the South Street per hour on an average?

300

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 300

***1 point***

15) Match the names of the street in Column A with the maximum and minimum number of vehicles expected to pass through the street on an average (per hour) in Column B and Column C, respectively; in Table M2W2GA1. ***1 point***

| | Name of the connecting street | | The maximum number of vehicles expected to pass through the street (per hour) | | The minimum number of vehicles expected to pass through the street (per hour) |
|---|---|---|---|---|---|
| | Column A | | Column B | | Column C |
| a) | Connecting West and North street | i) | 300 | 1) | 0 |
| b) | Connecting East and North street | ii) | 300 | 2) | 100 |
| c) | Connecting East and South street | iii) | 400 | 3) | 600 |
| d) | Connecting West and South street | iv) | 900 | 4) | 0 |

Table : M2W2GA1

- [ ] d → i → 2
- [ ] b → iii → 3.
- [x] b → iv → 3.
- [x] d → ii → 1.
- [x] a → iii → 2.
- [ ] a → iii → 4.
- [x] c → i → 4.

Yes, the answer is correct.

Score: 1

Accepted Answers:

b → iv → 3.
d → ii → 1.
a → iii → 2.
c → i → 4.

**NOTE FOR OPTION-2 IN QUESTION-3: [For elementary row operation of type-2 (scaling rows by a constant), the constant has to be non-zero. Scaling a row by zero is not permitted] Please answer this question keeping this in mind.**